ENGR 1451/2451: Exploratory Data Science

Simple Linear Regression
Slope Inference and Prediction

Prof. Paul W. Leu
University of Pittsburgh

Fall 2026

Today: From a Fitted Line to an Inference

Brief review of simple linear regression

Residuals, R^2, and diagnostic plots tell us what a fitted line describes—and what it misses.

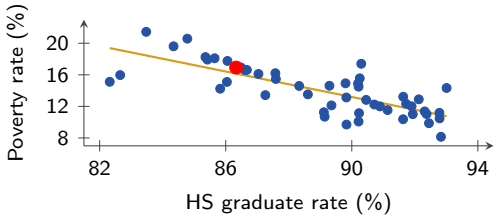
New question

A sample has a nonzero slope. Is that convincing evidence that the *population* slope is nonzero?

What we will learn

Check residuals, make inference for the slope, then compare **confidence intervals for the mean** and **prediction intervals for individuals**.

Review: What Did the State Regression Tell Us?



Fitted line

$$\widehat{y} \approx 85.789 - 0.807x$$

One residual

$e_i = y_i - \widehat{y}_i$

Georgia: $e_i \approx +0.77$ pp.

Two fit summaries

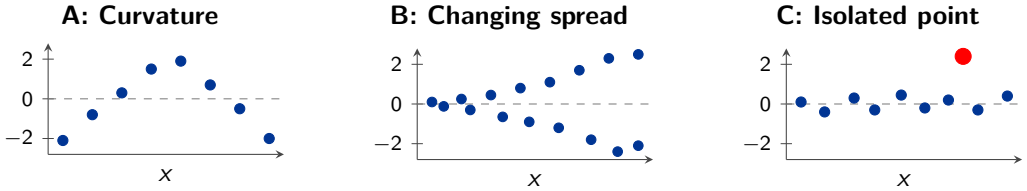
RMSE (typical residual size):

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n} e_i^2} \approx 1.90 \text{ pp}$$

$R^2 \approx 0.599$**:**

About 60% of the observed poverty-rate variation is accounted for by the line.

Review: What Do Residual Plots Diagnose?



Which plot questions the **straight-line mean**, which questions **constant variance**, and which calls for investigation of an **unusual observation**?

Describing a Sample Versus Inference

What we already know

$$\hat{y} = b_0 + b_1 x$$

b_1: **sample slope** computed from our observed data.

What we want to learn

$$E(Y \mid x) = \beta_0 + \beta_1 x$$

β_1: **population slope**, which we do not know.

A fitted line always exists

Even data with a weak or misleading pattern can produce a nonzero b_1.

Inference asks

Could sampling variation explain the observed b_1, if the true slope β_1 were zero?

Regression Model Assumptions

A model for the response at a fixed predictor value

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \qquad \varepsilon_i \overset{\text{ind}}{\sim} N(0, \sigma^2).$$

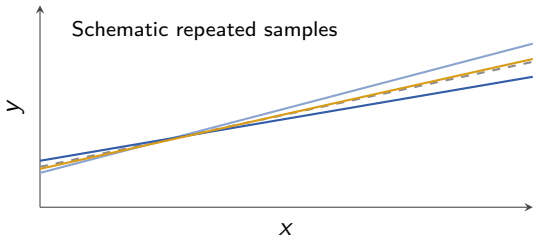
1. Straight-line mean
$E(Y_i \mid x_i) = \beta_0 + \beta_1 x_i$.

2. Independent observations
The errors for distinct observations are independent.

3. Roughly constant spread
Error SD σ does not depend on x.

4. Nearly Gaussian errors
Especially helpful for small samples.

What Is the Standard Error of a Slope?



Imagine repeating the study

Each random sample can give a slightly different fitted slope b_1.

Standard error

$SE(b_1)$ estimates the sample-to-sample spread of those slopes.

Key idea: A large $|b_1|$ matters more when $SE(b_1)$ is small.

Example: 27 Pairs of Identical Twins

Historical textbook dataset

x: IQ of the twin raised by biological parents.

y: IQ of the twin raised by foster parents.

$n = 27$ twin pairs; sample correlation $r = 0.882$.

Fitted model

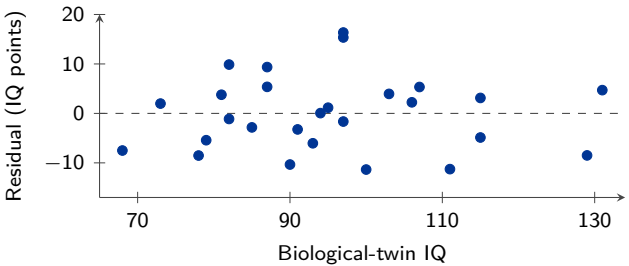
$$\widehat{\text{foster IQ}} = 9.2076 + 0.9014x$$

Questions

1. Is the population slope different from 0?
2. What slopes are compatible with the data?
3. Can we generalize beyond these 27 pairs?

$R^2 = 0.7779$ describes **in-sample fit**, not a causal effect.

Check the Twin Residuals Before Inference



Observed minus fitted:

$e_i = y_i - \widehat{y}_i$.

Look for:

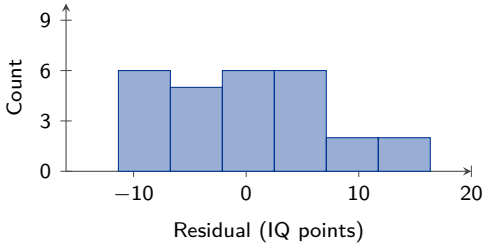
- Curvature.
- Changing vertical spread.
- Isolated observations.

Discuss: Which concerns do you see here?

```
plt.scatter(twins["biological"], model.resid)
plt.axhline(0, linestyle="--")
plt.show()
```

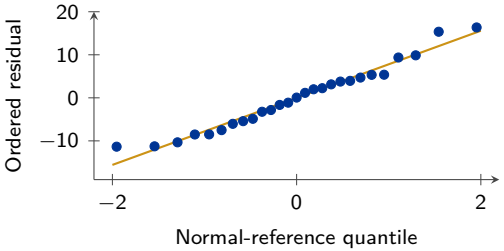
Check the Shape of the Residuals

Histogram



Look for asymmetry and extreme residuals.

Normal probability plot



Roughly straight points support a normal error shape.

Check residuals, not the raw x or y values. Neither plot establishes independence or representative sampling.

Sensitivity Check: Does One Pair Drive the Slope?

Refit without the observation with the largest absolute residual. Here it is the pair $(x, y) = (97, 113)$, with residual about $+16.35$ IQ points.

```
pair_to_check = model.resid.abs().idxmax()
model_without = smf.ols(
    "foster ~ biological", data=twins.drop(index=pair_to_check)
).fit()
```

Data used	n	Intercept	Slope
All pairs	27	9.2076	0.9014
Without selected pair	26	9.0068	0.8969

Does this pair materially change the fitted slope?

Discussion: Which Hypotheses Test the Slope?

We want to know whether the **population mean response** has a linear relationship with x.

Choose one pair of hypotheses

1. $H_0 : b_0 = 0, \quad H_A : b_0 \neq 0$
2. $H_0 : \beta_0 = 0, \quad H_A : \beta_0 \neq 0$
3. $H_0 : b_1 = 0, \quad H_A : b_1 \neq 0$
4. $H_0 : \beta_1 = 0, \quad H_A : \beta_1 \neq 0$

A t-Test for the Population Slope

Test statistic

$$T = \frac{b_1 - \beta_{1,0}}{SE(b_1)}, \qquad \boxed{df = n - 2}$$

For a no-trend null: $\beta_{1,0} = 0$.

A two-sided test compares $|T|$ to a t distribution with $n - 2$ degrees of freedom.

Why $n - 2$?

We fit **two coefficients**:

β_0 (intercept)

β_1 (slope).

For $n = 27$ twin pairs:

$df = 27 - 2 = 25$.

Read the Regression Table: Which Row Matters?

Model for 27 twin pairs

	Estimate	Std. Error	t value	Two-sided p
Intercept	9.2076	9.2999	0.99	0.3316
Biological twin IQ	**0.9014**	**0.0963**	**9.36**	$\mathbf{1.2 \times 10^{-9}}$

Estimate $b_1 = 0.9014$
One additional biological-twin IQ point is *associated* with about 0.90 additional fitted foster-twin IQ points.

Standard error $SE(b_1) = 0.0963$
How uncertain is the slope estimate under the sampling/model assumptions?

Discussion: What Does R^2 Really Mean?

The twin regression has $R^2 = 0.7779$ and $\widehat{y} = 9.2076 + 0.9014x$, where x is biological-twin IQ.

Which statement is *false*?

1. A 10-point increase in x corresponds to about 9 additional fitted foster-IQ points.
2. The line accurately predicts approximately 78% of *individual* foster-IQ scores.
3. The model can be written as $\widehat{y} \approx 9.2 + 0.90x$.
4. About 78% of the *observed variation* in foster IQ is accounted for by the fitted line.

Question adapted from OpenIntro Statistics (2nd ed.), Ch. 7 regression-output discussion (CC BY-SA).

Can We Test the y-Intercept Too?

Yes: use the same t logic

$$H_0 : \beta_0 = 0, \qquad H_A : \beta_0 \neq 0$$

$$T = \frac{b_0}{SE(b_0)}, \qquad df = n - 2$$

Twin regression output

$b_0 = 9.2076$, $SE(b_0) = 9.2999$

$t \approx 0.99$, $p \approx 0.332$.

But what does it ask?

Is the *mean foster IQ* at biological-twin IQ $x = 0$ different from zero?

The scientific question matters

Here the observed x range is 68–131, so $x = 0$ is extreme extrapolation.

In **sensor calibration**, testing a zero-concentration signal can be useful.

Work the Slope Test

Under $H_0 : \beta_1 = 0$

$$T = \frac{0.9014 - 0}{0.0963} = 9.36$$

$$df = 27 - 2 = 25$$

Two-sided p-value

$$p = P(|T_{25}| \geq 9.36)$$

$$\boxed{p \approx 1.2 \times 10^{-9}}$$

Conclusion—conditional on assumptions

The data provide extremely strong evidence against a zero *linear* population slope.

Statistical significance does **not** establish:

- Causality.
- Accurate predictions for individual twin pairs.

Discussion: Interpret a Very Small p-Value

The slope test reports $p \approx 1.2 \times 10^{-9}$.

Which interpretation is correct?

1. The probability that the population slope equals zero is about 1.2×10^{-9}.
2. If the true population slope were zero and the model held, a result at least this extreme would be extraordinarily unlikely.
3. The slope is so significant that biological IQ must causally determine foster IQ.

A Confidence Interval for the Slope

General 95% confidence interval

$$\boxed{b_1 \pm t^*_{df=n-2}\, SE(b_1)}$$

For 27 observations:

$df = 25$, and $t^*_{25} \approx 2.06$ for 95% confidence.

From the regression output

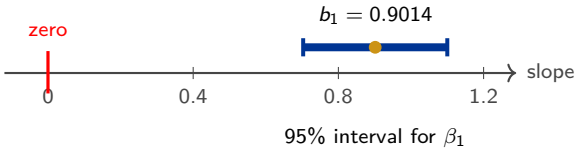
$$b_1 = 0.9014,$$
$$SE(b_1) = 0.0963.$$

Margin of error

$2.06(0.0963) \approx 0.198.$

Calculate and Interpret the 95% Interval

$$0.9014 \pm 2.06(0.0963) \approx \boxed{(0.703,\ 1.100)}$$



In context

Under the assumptions, plausible population slopes are about 0.70 to 1.10 foster-IQ points per biological-IQ point.

Connection: The interval excludes 0, agreeing with rejection of $H_0 : \beta_1 = 0$ by a two-sided 5% test.

Three Different Intervals in Regression

1. Confidence interval for the slope β_1

A *single range of plausible population slopes*: 0.703 to 1.100 IQ points per IQ point for our example.

2. Confidence interval for the mean at x_0

How uncertain is the **population average** $E(Y \mid x_0)$?

Plot many such intervals to form a **confidence band** around the line.

3. Prediction interval at x_0

Where might **one new response** Y_{new} fall?

This band is **wider**: individuals vary around the mean line.

Confidence Interval vs. Prediction Interval

At a chosen predictor value x_0, the fitted line gives

$$\boxed{\widehat{y}_0 = b_0 + b_1 x_0}.$$

Confidence interval: population mean response

What is the **average response** when $x = x_0$?

This measures uncertainty in the estimated height of the mean line.

Prediction interval: one new response

Where might **one new observation** fall when $x = x_0$?

This also includes individual scatter around the mean line.

Same center, different uncertainty: the prediction interval is wider.

Understanding the Terms in the Formulas

Symbol	Meaning
n	Number of observations in the sample
x_0	Predictor value of interest
$\bar{x}$	Average predictor value in the sample
$\hat{y}_0$	Predicted response from the fitted line at x_0
$e_i = y_i - \hat{y}_i$	Residual: observed minus predicted response
$s = \sqrt{\frac{\sum_i e_i^2}{n-2}}$	Residual standard deviation: typical vertical scatter around the fitted line
$S_{xx} = \sum_i (x_i - \bar{x})^2$	Total squared spread of the predictor values
t^*	Critical value from the t-distribution; for a 95% interval, $t^* = t_{0.975,n-2}$

Confidence Interval for the Mean Response

How precisely can we estimate the **mean response** at x_0?

95% confidence interval

$$\hat{y}_0 \pm t^* s \sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}$$

What determines the interval width?

- s**:** More scatter around the fitted line makes the interval wider.
- $1/n$**:** More observations generally reduce uncertainty.
- $(x_0 - \bar{x})^2/S_{xx}$**:** Predictions farther from the average predictor value $\bar{x}$ are less precise.

The confidence interval is narrowest at $x_0 = \bar{x}$.

Prediction Interval for One New Observation

A new observation can fall above or below the line, even when the mean is estimated precisely.

95% prediction interval

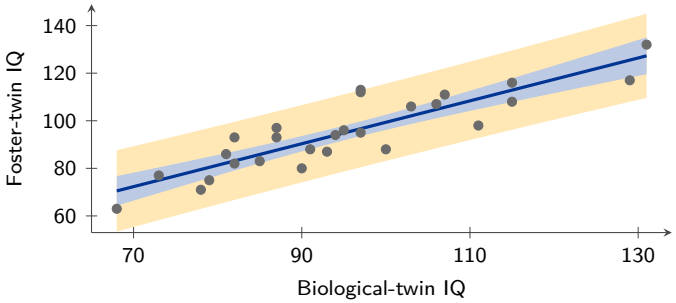
$$\hat{y}_0 \pm t^* s \sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}$$

Why the extra 1?

- The mean CI includes uncertainty in the estimated mean.
- The PI also includes **random scatter of one new observation**.
- The extra 1 represents this additional source of variation.

More data can locate the mean more precisely; individual scatter remains.

See the Difference: Mean CI vs. Individual PI



At $x_0 = 100$

Fitted mean:

$\widehat{y}_0 = 99.35$

Mean 95% CI

[96.15, 102.55]

Individual 95% PI

[83.11, 115.59]

Mean CI

Individual PI

Discussion: Which Interval Answers the Question?

Assume the regression model is reasonable for predicting foster IQ from biological-twin IQ at $x_0 = 100$.

Choose a 95% mean confidence interval or a 95% prediction interval

1. We want to estimate the average foster IQ among people with $x_0 = 100$.
2. We want a range for the foster IQ of one additional twin with $x_0 = 100$.
3. A researcher wants an interval for the population slope, regardless of x_0.

Small p-Value, But Can We Trust the Inference?

Model conditions

- Residual curvature $\rightarrow$ mean may not be linear.
- Fan shape $\rightarrow$ variance is not constant.
- Extreme residuals $\rightarrow$ assess normality and influence.

Study design

- Are observations independent?
- Are cases randomly sampled or representative?
- What population is being inferred about?

An extremely small computed p-value cannot fix biased selection or a misspecified model.

Discussion: Should We Test the 51 State Slopes?

Return to our states poverty example

Our plotted observations cover the **50 states and DC** (2017 aggregates). We did not randomly sample states from a larger collection of states.

Discuss

1. What population or random process would our slope test refer to?
2. What can we still report *without* a p-value?

Python: Slope Inference with `ols`

Use the model fitted in the notebook

```
import statsmodels.formula.api as smf

model = smf.ols(
    "foster ~ biological", data=twins
).fit()

print(model.params["biological"])
print(model.bse["biological"])
print(model.pvalues["biological"])
print(model.conf_int())
```

What the output means

`params`: coefficient estimates.

`bse`: coefficient standard errors.

`pvalues`: two-sided tests against zero.

`conf_int()`: 95% coefficient intervals.

`smodel.resid` supplies the residual checks; `get_prediction(...).summary_frame()` supplies mean and prediction intervals.

Takeaways: Three Different Inferences

1. **Coefficient inference**: we can test or form CIs for both β_1 (slope) and β_0 (intercept). The slope is usually more scientifically relevant.

2. **Slope CI**: $b_1 \pm t^* SE(b_1)$ estimates the population slope—not the location of an individual response.

3. **Mean CI vs. prediction interval**: a mean-response CI describes the population average at x_0; a **wider prediction interval** describes one new Y.

4. **Validity**: check residual patterns, shape, sensitivity to unusual observations, and study design. R^2 and small p-values do not establish causation or guarantee individual accuracy.